

PROJECT 4

Pune Real Estate Market Trends

Maharashtra RERA Open Data Analysis • Google Sheets

1,932	340	₹66.5L
RERA Registered Projects	Property Transactions	Avg Deal Value
800	721	401
Active Projects (41.4%)	Completed (37.3%)	Lapsed (20.8%)

1. Project Overview

This project is the fourth in a data analysis series focused on building practical skills in Google Sheets through real-world, publicly available datasets. Project 4 graduates from synthetic or cleaned teaching data to genuine government open data — requiring data cleaning, interpretation, and domain understanding before any analysis can begin.

The dataset was sourced from the Maharashtra Real Estate Regulatory Authority (RERA) portal, a government body established under the Real Estate (Regulation and Development) Act, 2016. RERA mandates that all residential real estate projects above a certain threshold register with the authority, providing a comprehensive and legally mandated record of the market.

1.1 Datasets Used

Dataset	Source	Records	Scope
RERA Project Registry	Maharashtra RERA Open Data	1,932 projects	All of Pune district
Property Transactions	RERA Transaction Data	340 transactions	Erandwane locality

1.2 Project Objectives

- Analyse the overall health and status distribution of RERA-registered projects in Pune
- Track how project registrations have evolved year-over-year since RERA's inception
- Identify high-activity localities and understand geographic concentration
- Examine real property transaction data: deal types, price distributions, and market segments

- Build and document six distinct chart types in Google Sheets
- Practise working with messy, real-government data including inconsistent formatting and missing values

2. Dataset Analysis

2.1 Project Status Distribution

Of the 1,932 RERA-registered projects in Pune, the majority remain active, indicating a market with substantial ongoing construction activity. A meaningful share has lapsed — projects that exceeded their RERA-approved completion deadline without renewal — which is a notable finding with real implications for home buyers.

Status	Count	Percentage	Interpretation
Active	800	41.4%	Under construction or within approved timeline
Completed	721	37.3%	Handed over / occupancy certificate obtained
Lapsed	401	20.8%	Deadline exceeded without renewal — buyer risk
De-registered	9	0.5%	Cancelled / removed from registry
Total	1,932	100%	

Key insight: The 20.8% lapse rate is significant. It means roughly 1 in 5 registered projects missed their RERA deadline — a pattern that RERA itself monitors and penalises. For data analysis purposes, lapsed projects needed to be handled carefully as they are not failures in the conventional sense; many eventually renew or complete outside the tracked window.

2.2 Year-on-Year Registration Trend

RERA was enacted in 2016 and the Maharashtra RERA portal opened for registrations in 2017. This created a landmark spike in that year as developers rushed to comply with the new law before its enforcement deadline.

Year	Projects Registered	Notes
2017	412	Peak year — compliance rush at RERA launch
2018	Significant	Post-launch normalisation
2019	Moderate	Stable market activity
2020	Lower	COVID-19 impact on registrations
2021–23	Recovering	Post-pandemic recovery trajectory

Peak registration year: 2017 with 412 projects. This is not a market boom year per se, but a regulatory compliance event. Separating regulatory artefacts from genuine market signals is an important data literacy skill practised in this project.

2.3 Top Localities by Project Count

Four localities dominate project registrations, reflecting Pune's geographic expansion over the RERA period. Kothrud leads as a mature, established western suburb; Punawale and Kharadi represent newer high-growth corridors; Erandwane is a premium central locality.

Rank	Locality	Projects	Market Character
1	Kothrud	166	Established western suburb, mid to premium segment
2	Punawale	116	Emerging corridor, affordable to mid segment
3	Kharadi	100	IT hub adjacency, strong investor demand
4	Erandwane	80	Premium central locality, high land value

3. Property Transaction Analysis (Erandwane)

The 340-row transaction dataset covers property deals registered in Erandwane, providing a granular view of one of Pune's most sought-after residential areas. Each row represents a single registered property transaction with details on deal type, floor, area, and agreed value.

3.1 Transaction Types

The most common transaction type is Agreement to Sell (ATS), which is a pre-possession agreement between builder and buyer. ATS documents are registered at the time of booking, making them the dominant transaction type in an under-construction-heavy market like Pune's.

Transaction Type	Count	Description
Agreement to Sell (ATS)	61	Most common — pre-possession builder-buyer agreement
Sale Deed	Various	Post-possession transfer of ownership
Others	Remaining	Gift deeds, mortgage releases, etc.

3.2 Market Value Summary

₹66.5L	₹41.3L	₹25L–1Cr
Average Deal Value	Median Deal Value	Most Active Price Band

The gap between average (₹66.5L) and median (₹41.3L) deal value indicates a right-skewed distribution — a smaller number of high-value transactions (luxury apartments, large floor plates) are pulling the average upward while the bulk of deals cluster in the ₹25L–50L range. This is characteristic of a market with a mix of affordable apartments and premium properties.

3.3 Price Segment Distribution

Value segmentation was performed by creating a custom column using nested IF formulas to categorise each transaction into a price band. This technique allows analysis of market depth at different price points.

Price Segment	Range	Volume	Market Insight
Budget	Below ₹25L	Lower	Typically small units / older stock
Mid Market	₹25L – ₹50L	Highest	Dominant segment — core Erandwane buyer
Upper Mid	₹50L – ₹1Cr	Significant	2–3 BHK premium apartments
Premium	Above ₹1Cr	Lower	Luxury / penthouse / large configurations

4. Charts Created

Six charts were built in Google Sheets to visualise different facets of the dataset. Each chart type was selected to best represent its underlying data structure, and each introduced a new Google Sheets skill.

#	Chart	Type	Key Skill Practised
1	Project Status Distribution	Donut Chart	Donut chart creation, data labels with percentages
2	Year-on-Year Registration Trend	Stacked Column Chart	Stacked columns, secondary data series, trend annotation

#	Chart	Type	Key Skill Practised
3	Top Localities Comparison	Grouped Bar Chart	Grouped bars, horizontal orientation, comparative layout
4	Transaction Types Breakdown	Horizontal Bar Chart	Sorting by frequency, data labels on bars
5	Market Value Segments	Column Chart	Value segmentation via IF formula, segment colouring
6	Floor-wise Transaction Distribution	Column / Bar Chart	Floor extraction from raw text, frequency aggregation

5. New Skills Acquired

This project was designed to push beyond the charting techniques covered in Projects 1–3. Each new skill listed below was deliberately introduced through the needs of the real dataset.

5.1 Google Sheets Charting Techniques

- Donut Chart: Creating a ring chart with a hollow centre, applying percentage data labels directly on segments, and formatting label colour for readability against coloured segments
- Stacked Column Chart: Layering multiple data series in a single column to show composition over time, with a correctly ordered legend
- Grouped Bar Chart: Placing multiple series side-by-side for direct comparison across categories (e.g. locality vs project count by status)
- Data Labels on Charts: Adding value labels directly to chart elements, controlling font size, position (inside end, outside end, centre), and number formatting
- Alternating Row Colours: Applying banded shading to data tables using Format → Alternating Colours for readability in large datasets

5.2 Data Analysis Techniques

- Value Segmentation: Deriving categorical segments from continuous numeric data using nested IF() or IFS() formulas to create price band labels
- COUNTIF across segments: Aggregating transaction counts by derived segment using COUNTIF with both cell references and wildcard criteria
- Real Government Open Data Analysis: Handling messy, inconsistently formatted records from a live government portal (missing values, mixed date formats, duplicate entries, encoding issues)
- Regulatory Data Interpretation: Understanding that data artefacts (e.g., the 2017 spike) can reflect regulatory events rather than market fundamentals
- Distribution Analysis: Comparing mean vs median to detect skewness and outliers in deal value data

5.3 Data Cleaning Practices

- Trimming whitespace from locality names that caused duplicate groupings in COUNTIF
- Standardising transaction type labels (mixed capitalisation, abbreviations)

- Handling blank cells in the registration year column without breaking COUNTIF formulas
- Extracting floor numbers from a free-text description field using LEFT, MID, FIND combinations

6. Key Findings & Insights

6.1 Market Health

- Pune's RERA-registered market is predominantly active (41.4%), suggesting a live construction pipeline rather than a saturated or declining market
- The 20.8% lapse rate warrants attention: one in five projects has exceeded its registered completion date, which may indicate developer financial stress, construction delays, or regulatory non-compliance
- The low de-registration count (9 projects, 0.5%) suggests RERA enforcement has not yet led to wide-scale project cancellations

6.2 Geographic Concentration

- The top 4 localities (Kothrud, Punawale, Kharadi, Erandwane) account for a disproportionate share of total registered projects, reflecting Pune's radial growth pattern from the city core
- Punawale's high rank despite being a newer locality indicates the westward expansion corridor's strong developer activity
- Erandwane's 80 projects despite its premium pricing confirms sustained demand in established central neighbourhoods

6.3 Transaction Market

- Agreement to Sell dominates (61 of sampled transactions), confirming that Erandwane buyers are primarily engaging in under-construction purchases rather than secondary market resale
- The mean-median spread (₹66.5L vs ₹41.3L) reveals a long right tail: a few high-value deals significantly lift the average, making median the more representative measure for typical buyer experience
- The most active price band (₹25L–1Cr) encompasses the majority of deals, spanning from compact 1 BHK units to spacious 2–3 BHK configurations

7. Tools & Methods

Component	Detail
Primary Tool	Google Sheets (browser-based)
Data Source	Maharashtra Real Estate Regulatory Authority (RERA) — Government Open Data Portal
Dataset 1	RERA Project Registry — 1,932 Pune projects
Dataset 2	Property Transactions — 340 records, Erandwane
Formulas Used	COUNTIF, IFS, IF, TRIM, LEFT, MID, FIND, AVERAGE, MEDIAN
Chart Types	Donut, Stacked Column, Grouped Bar, Horizontal Bar, Segmented Column

Component	Detail
Formatting	Alternating row colours, conditional number formatting, data labels
Series	Project 4 of ongoing Google Sheets Data Analysis series

This project used real, unmodified government data from the Maharashtra RERA portal. All analysis, charts, and insights were produced entirely in Google Sheets.